



Assignment_AREA_Mensuration

Exercise

1. Find the area of a triangle whose base is 32 cm and altitude is 36 cms. **Ans: 576 cm²**
2. Find the altitude of a triangle whose base is 38 cm and area is 855 cm². **Ans: 45 cm**
3. Find the area of a triangle whose base is 45 cm and altitude is 40 cms. **Ans: 900 cm²**
4. Find the base of a triangle whose area is 864 cm² and altitude is 48 cms. **Ans: 36 cm**
5. If the area of a triangle is 500 cm² and base: height is 5:8, find its height and base. **Ans: 25 cm, 40cm**
6. If the area of a triangle is 320 cm² and base: height is 5:8, find its height and base. **Ans: 20 cm, 32cm**
7. Two sides of the triangle are 15 cm and 18 cm. If height of the triangle corresponding to 18 cm sides is 24 cm find-
 - a) area of the triangle. **Ans: 216 cm²**
 - b) height of the triangle corresponding to 15 cm side. **Ans : 28.8 cm**
8. Two sides of the triangle are 12 cm and 14 cm. If height of the triangle corresponding to 14 cm sides is 18 cm find-
 - a) area of the triangle. **Ans: 126 cm²**
 - b) height of the triangle corresponding to 12 cm side. **Ans : 21 cm**
9. If the base and height of a triangle are in the ratio of 4:5 and area is 490 cm². Find its base and height. **Ans: 28 cm and 35 cm.**
10. If the base and height of a triangle are in the ratio of 2:5 and area is 125 cm². Find its base and height. **Ans: 10 cm and 25 cm**
11. In a right-angled triangle, the length of two legs are 16 and 12 cms. Find the length of hypotenuse and its area. **Ans: 20 cm² and 96 cm²**
12. In a right-angled triangle, the area is 60 cm² and one of its legs is 8 metres. Find its other leg and hypotenuse. **Ans: 15m, 17m**

Assignment_AREA_Mensuration

13. In two triangles, the ratio of the areas is 6:5 and the ratio of their heights is 5:6. Find the ratio of their bases.
Ans: 36:25

14. The base of a triangular field is 800 metres and its height 500 metres. Find the area of the field and also calculate the charges for supplying water to the field at the rate of Rs. 30 per sq. hectare.
A: 200000 m², Rs. 600 (one hect. = 10,000 m²)

15. The sides of an equilateral triangle are 6m. find its area.

Ans: $9\sqrt{3} \text{ m}^2$

16. If the perimeter of an equilateral triangle is 24 meter, find its area.

Ans:

$16\sqrt{3} \text{ m}^2$

17. The perimeter of an isosceles triangle is 300 cm, if the base is 60 cm, find the length of equal sides. **Ans:**

120 cm

18. Find the area of a right-angled triangle whose base is 20 cm and hypotenuse 25 cm.

Ans: 150

cm²

19. If the side of an equilateral triangle is 8 cm, find its area and perimeter.

Ans:

$16\sqrt{3} \text{ cm}^2$, 24 cm.

20. The length of the sides of a triangle are in the ratio of 4:5:6 and its perimeter is 60 cm. Find its area.

Ans: $60\sqrt{7} \text{ cm}^2$

21. One side of a rectangular field is 20 m and one of its diagonals is 25 m. Find the area of the field. **Ans:**

300 m²

22. The length of a rectangular field is 3 times its width. If the perimeter of the field is 24 m, then find the area of the field. **Ans: 27 m²**

23. If the perimeter of a rectangular field is 200 m and its breadth is 40 m. Find its area. **Ans: 2400 m².**

24. If the perimeter of a square is 16 cm. Find its area.

Ans: 16 cm².

25. The length of a rectangle is 1 cm more than its width. Its perimeter is 14 cm. Find the area of rectangle.

Ans: 12 cm².

26. A rectangular plot has its sides in the ratio 4:5. The area of the lawn is $\frac{1}{15}$ hectares. Find the length and breadth of the lawn.

Ans: $40/\sqrt{3} \text{ m}$, $50/\sqrt{3} \text{ m}$

27. The length of a rectangular plot is 75% more than its breadth. If the difference between length and breadth is 30 cm, find the area of the rectangle. **Ans: 2800 cm²**

Assignment_AREA_Mensuration

28. The length of a rectangle is four times its breadth. If its length is increased by 10 cm. and breadth is increased by 10 cm, the area of the rectangle is increased by 600 cm^2 . Find the length of the rectangle.

Ans: 40 cm.

29. How many meters of carpet 5m wide will be required to cover the floor of a room which is 25 m long and 14 m broad.

Ans. 70m

30. Find the area of the largest circle that can be drawn in a rectangle of sides 28 cm and 34 cm.

Ans: 616 cm^2

31. The area of a rectangle is 560 square metres. If the length is 40% more than the breadth, what is the breadth of the rectangular field?

Ans: 20 m

32. The ratio between the perimeter and the breadth of a rectangular is 7:1. If the area of the rectangle is 360 sq. cm, what is the length of the rectangle?

Ans: 30 cm

33. The area of rhombus is 210 cm^2 . The length of one of its diagonals is 14 cm. The length of the other diagonal is:

Ans: 30 cm

34. The area of rhombus is 320 cm^2 . The length of one of its diagonals is 16 cm. The length of the other diagonal is:

Ans: 40 cm

35. If the diagonals of a rhombus are 16 cm and 12 cm, the area and the perimeter of the rhombus are respectively.

Ans: 96 cm^2 , 40 cm

36. The radius of a circle is 14 metres. Find its area and circumference.

Ans: 616 m^2 ,

88 m

37. A rectangular lawn $80\text{m} \times 60\text{m}$ has two roads each 4 metres wide, running in the middle of it, one parallel to the length and the other parallel to the breadth. Find the cost of gravelling the road at the rate of Rs 12 per square metre.

Ans: Rs 6,528

38. The area of a circle is 3850 m^2 . Find its diameter and circumference.

Ans: 70m,

220 m

39. The area of a circle is 616 m^2 . Find its diameter and circumference.

Ans: 28m,

88 m

40. A circular grass lawn of 21 metres in radius has a path 7 metres wide running around it on the outside. Find the area of the path.

Ans: 1078 m^2

41. A circular grass lawn of 49 metres in radius has a path 7 metres wide running around it on the outside. Find the area of the path.

Ans: 2310 m^2

Assignment_AREA_Mensuration

42. The difference between the circumference and radius of a circle is 148 metres. Find its circumference and radius.

Ans: 176 m, 28 m

43. The difference between the circumference and radius of a circle is 111 metres. Find its circumference and radius.

Ans: 132 m, 21 m

44. One side of a parallelogram is 30 cm and its distance from the opposite side is 9 cm. The area of the parallelogram is:

Ans: 270 cm²

45. One side of a parallelogram is 20 cm and its distance from the opposite side is 8 cm. The area of the parallelogram is:

Ans: 160 cm²

46. From a square cloth a circle of largest possible area was cut out. If the area of the circle is 1386 cm², find the original area of the cloth.

Ans: 1,764 cm²

47. From a rectangular wooden board of 250 cm x 126 cm, a carpenter have cut out the largest possible circle.

a) Find the area of the circle. **Ans: 12,474 cm²**

b) Find the area of wasted wooden board. **Ans: 19,026 cm²**

48. 2 metres broad pathway is to be constructed around a rectangular plot on the inside. The area of the plot is 320 sq. m and length of the plot is 20 m. The rate of construction is Rs. 125 per square metre. Find the total cost of the construction.

Ans: Rs. 16,000

49. A rectangular lawn of 15 m length and 10 m width has a gravel path outside all around it. If the area of the gravelled pathway is 54 sq. m., what is the width of the path?

Ans: 1 m.

50. A rectangular lawn of 23 m length and 12 m width has a path of 2.5 m wide all around it on the inside. Find the area of the path and also the cost of constructing the path at rate of Rs. 33 per sq. mtrs.

Ans: 150 m², Rs. 4,950.